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Overview

Diabetes affects millions of people worldwide, and early detection can significantly improve patient outcomes. In this analysis, we explore lifestyle and health indicators to understand which factors are most predictive of diabetes. Using the Behavioral Risk Factor Surveillance System (BRFSS) dataset for each given year, we aim to classify individuals by diabetes status and uncover insights that can guide preventive healthcare strategies.

Objective

Build a classification model to predict diabetes risk and identify key health and lifestyle indicators that influence outcomes.

Architecture / Project Structure

Source Folder

- [Deliverable 1](#)
 - [Outputs/](#) - Contains generated machine learning notebooks for each year analyzed
 - [Master_Notebook.ipynb](#) - Orchestrator notebook that runs the full pipeline and launches the frontend UI
 - [Frontend.ipynb / Frontend.py](#) - Code for the Streamlit-based user interface
 - [Input.ipynb](#) - Parameterized notebook used to train and evaluate models for each selected year
- [Deliverable 2](#) - Video showcasing UI and walkthrough through sample year (2023)
- [Medium Blog Post](#) - Written walkthrough of the 2023 analysis (mirrors Deliverable 2)
- [data/](#) - Contains all source data and generated assets used by the UI

Instructions to Run Deliverable 1

1. Create shortcut to folder to get project path to 'MyDrive'
 - a. Right click the [Source Folder](#) -> Organize -> Add Shortcut -> All Locations -> MyDrive
2. Navigate to [Master_Notebook.ipynb](#)
3. Run *only* very [last cell](#) in [Master_Notebook.ipynb](#)
4. After running the last cell, a Streamlit URL will be generated in the output
5. Open the provided link in your browser

```
Mounted at /content/drive
Executing: 100% [4/4] [00:04<00:00, 1.01cells/s]
2026-04-15T19:36:53Z INF Thank you for trying Cloudflare Tunnel. Doing so, without a Cloudflare account, is a quick way to experiment and try it out. However, be aware that these accounts are not intended for production use.
2026-04-15T19:36:53Z INF Requesting new quick Tunnel on trycloudflare.com...
2026-04-15T19:36:56Z INF Your quick Tunnel has been created! Visit it at (it may take some time to be reachable):
2026-04-15T19:36:56Z INF https://explicitly-ant-optional-exams.trycloudflare.com
2026-04-15T19:36:56Z INF Cannot determine default configuration path. No file [config.yml config.yaml] in [~/cloudflared ~/cloudflared-warp ~/cloudflared-warp /etc/cloudflared /usr/local/bin]
2026-04-15T19:36:56Z INF Version 2026.3.0 (Checksum 4a9e50e6d6d798e90fcd01933151a90bf7edd99a0a55c28ad18f2e16263a5c30)
2026-04-15T19:36:56Z INF OS: Linux, GOVersion: go1.24.13, GOArch: amd64
2026-04-15T19:36:56Z INF Settings: map[ha-connections:1 protocol:quic url:http://localhost:8501]
2026-04-15T19:36:56Z INF cloudflared will not automatically update when run from the shell. To enable auto-updates, run cloudflared as a service: https://developers.cloudflare.com/cloudflared/
2026-04-15T19:36:56Z INF Generated Connector ID: b4986857-036d-4797-8d5d-2d2c401d9a97
2026-04-15T19:36:56Z INF Initial protocol quic
2026-04-15T19:36:56Z INF ICMP proxy will use 172.28.0.12 as source for IPv4
2026-04-15T19:36:56Z INF ICMP proxy will use ::1 in zone lo as source for IPv6
2026-04-15T19:36:56Z INF ICMP proxy will use 172.28.0.12 as source for IPv4
2026-04-15T19:36:56Z INF ICMP proxy will use ::1 in zone lo as source for IPv6
2026-04-15T19:36:56Z INF Starting metrics server on 127.0.0.1:20241/metrics
2026-04-15T19:36:56Z INF Tunnel connection curve preferences: [X25519MLKEW768 CurveP256] connIndex=0 event=0 ip=198.41.200.193
2026-04-15T19:36:56Z INF Tunnel connection curve preferences: [X25519MLKEW768 CurveP256] connIndex=0 event=0 ip=198.41.200.193 location=atl08 protocol=quic
2026-04-15T19:36:56Z INF Registered tunnel connection connIndex=0 connections=39c50493-58dc-4a5d-b6bf-ee7841024e2c event=0 ip=198.41.200.193 location=atl08 protocol=quic
```